

PROJECT DOCUMENTATION · CONFIDENTIAL

Bullion Infrastructure

Active-active multi-cloud Kubernetes platform — running AWS EKS and GCP GKE as a single, mesh-joined cluster with cross-cloud load balancing, delivered entirely through infrastructure-as-code and CI/CD.

✓ Active-active cross-cloud mesh — LIVE & verified (traffic balances EKS ⇄ GKE)

Repository: Olisaemeka111/bullion-infrastructure · Version 1.0 · 2026-06-23

AWS/EKS

GCP/GKE

Mesh

CI/CD

Security

Observability

Data

Roadmap

0 Document control & contents

FIELD	VALUE
Title	Bullion Infrastructure — Project Documentation
Version / date	1.0 · 2026-06-23
Status	Operational — active-active mesh live; data layer + control-plane on roadmap
Audience	Engineering leadership, platform/SRE teams, security, auditors
Owner	Platform Engineering
Source of truth	Git repo (Terraform, workflows, mesh manifests, this document)

Contents

1	Executive summary	\$1
2	Business context & objectives	\$2
3	Solution architecture	\$3
4	Infrastructure components	\$4
5	Service mesh (Istio multi-primary)	\$5
6	CI/CD & automation	\$6
7	Security & compliance	\$7
8	Observability	\$8
9	Data layer (CockroachDB — planned)	\$9
10	Reference control plane (offline model)	\$10
11	Operations runbook	\$11
12	Testing & verification	\$12
13	Implementation journey	\$13
14	Current state & roadmap	\$14
15	Cost	\$15
16	Risks & recommendations	\$16
17	Appendices	\$17

1 Executive summary

What this is. Bullion Infrastructure is a production-style platform that runs Kubernetes in two public clouds — Amazon EKS and Google GKE — and makes them behave as **one active-active cluster**. A single application, deployed to both clouds, is served as one global service with traffic load-balanced across both providers simultaneously; if one cloud degrades, traffic automatically shifts to the other.

Why it matters. It removes single-cloud dependency: the platform survives a whole-cloud outage, avoids vendor lock-in, and lets workloads run close to users in multiple regions — without operators managing failover by hand.

2

CLOUDS, ACTIVE-ACTIVE

4

WORKER NODES (2/CLOUD)

100%

IAC + CI/CD DEPLOYED

0

STATIC CLOUD KEYS (OIDC)

Outcome (verified). The clusters are joined into one Istio mesh and load balancing across clouds is confirmed: 20 requests from the AWS cluster were served **12 by AWS and 8 by GCP**. The whole platform is reproducible from Git and deployed through gated GitHub Actions pipelines with keyless cloud authentication.

Delivered

- Two managed Kubernetes clusters (EKS + GKE), provisioned by Terraform.
- An Istio **multi-primary** service mesh joining both clusters across clouds (mTLS).
- **Active-active load balancing** of a global service across both clouds, with automatic failover (outlier ejection).
- A **cross-cloud private network** (AWS↔GCP HA VPN + BGP, 4 tunnels) — live.
- A **central distributed database** (CockroachDB multi-region, 4 nodes across both clouds over the VPN), deployed via CI/CD.
- A real **global active-active application** (the board game) built from source to GHCR and served from both clouds behind public ingress LoadBalancers.
- Full **CI/CD** (GitHub Actions) with keyless OIDC auth and approval gates.
- Centralised **observability** (Prometheus/Grafana).

On the roadmap

- In-cluster control-plane/dashboard (the agentic reconciler running on-cluster).
- A single global DNS endpoint (one hostname) fronting both ingresses — optional.
- Production hardening: CockroachDB TLS, least-privilege IAM, private GKE endpoint.

Commercial note. The environment currently runs on a GCP free trial and a pay-as-you-go AWS account for validation. Production hardening (least-privilege IAM, private API endpoints, quota uplifts, the data layer) is scoped in \$14/\$16.

2 Business context & objectives

Objectives

- **Resilience:** survive a full cloud-provider outage.
- **No lock-in:** one platform, portable across clouds.
- **Active-active:** use all capacity, not idle standby.
- **Automation:** everything reproducible via IaC + CI/CD.
- **Security by default:** keyless auth, mTLS, gated changes.

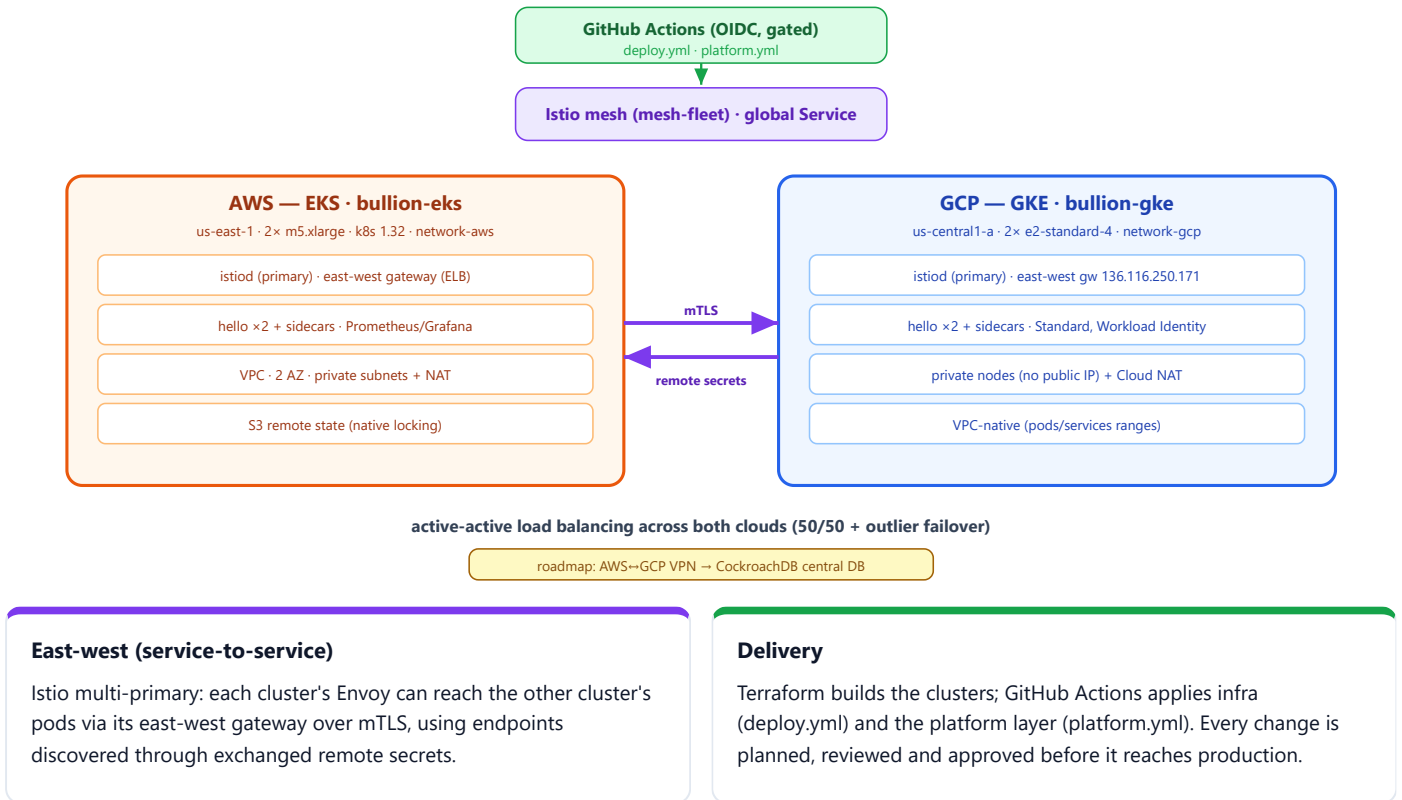
Non-goals (this phase)

- Production-grade least-privilege IAM (currently broad for validation).
- Stateful workloads at scale (data layer is on the roadmap).
- Azure (no subscription available; cleanly disabled, re-enableable).

Design principles

PRINCIPLE	HOW IT'S REALISED
Two control planes, one fabric	Each cluster keeps its own K8s + istiod; the mesh + global LB make them one logical platform (you never stretch one control plane across clouds).
Declarative & reproducible	Terraform for infra; manifests for the mesh; GitHub Actions to apply — all in Git.
Keyless & gated	GitHub OIDC to each cloud; production environment requires human approval.
Active-active	Global service spans both clusters; DestinationRule splits 50/50; outlier detection fails over.

3 Solution architecture



AWS EKS — bullion-eks

Account / region	730767193290 / us-east-1
Kubernetes	1.32 (managed control plane)
Nodes	2 × m5.xlarge (4 vCPU/16GB), on-demand
Networking	VPC, 2 AZ, private node subnets + single NAT, public subnets, EKS subnet tags
Access	EKS access entries (CI role + operator = cluster-admin)
Mesh ports	node SG opened for istiod webhook 15017 + xDS 15012

GCP GKE — bullion-gke

Project / zone	project-f38bfcd6-70ff-4a8d-9cd / us-central1-a
Mode	GKE Standard (zonal), Workload Identity enabled
Nodes	2 × e2-standard-4 (4 vCPU/16GB)
Networking	VPC-native; private nodes (no public IP) + Cloud NAT egress
Control plane	public endpoint (open) for CI/operator access
Constraint	private nodes chosen to fit free-trial IP quota

Shared / cross-cutting

COMPONENT	DETAIL
IaC	Terraform ≥ 1.10, modules: <code>eks</code> , <code>gke</code> , <code>global_dns</code> , <code>cross_cloud</code>
Remote state	S3 bucket <code>fleet-mini-tfstate-59998ac4</code> , native state locking (no DynamoDB)
Naming	<code>bullion-*</code> prefix across clusters + networks
Azure (AKS)	module present but commented out; <code>enable_azure=false</code> (no subscription)

5 Service mesh — Istio multi-primary

Both clusters form one mesh (`meshID = mesh-fleet`). Multi-primary = an independent istiod per cluster; multi-network = traffic crosses clouds via east-west gateways.

ELEMENT	IMPLEMENTATION
Trust	Shared root CA issues each cluster's istiod intermediate (the <code>cacerts</code> secret) → cross-cloud mTLS works.
Control plane	One istiod per cluster (no single point of failure).
Cross-cluster transport	East-west gateways (LoadBalancer): EKS ELB hostname; GKE IP 136.116.250.171.
Discovery	Remote secrets (<code>istio-remote-secret-eks/-gke</code>) let each istiod read the other's endpoints.
Network identity	<code>topology.istio.io/network</code> on istio-system (network-aws / network-gcp) — required for cross-network routing.
Global service	<code>hello.demo.svc.cluster.local</code> deployed to both; <code>DestinationRule</code> 50/50 cross-region + <code>outlierDetection</code> failover.

Verification. 20 calls from an EKS client pod → **12 AWS / 8 GCP**. Cross-cloud active-active load balancing confirmed.

6 CI/CD & automation

PIPELINE	RESPONSIBILITY	TRIGGER / GATE
<code>ci.yml</code>	80 unit/integration tests (stdlib) + <code>terraform fmt / validate</code>	every push/PR · no cloud creds
<code>deploy.yml</code>	Terraform infra (EKS, GKE, VPN): plan → apply → destroy	PR=plan · merge=apply (gated) · dispatch=destroy
<code>platform.yml</code>	Istio join + global app + observability across both clusters	dispatch / mesh changes (gated)

Flow

Open PR → automated **plan** posted as a comment → review → merge → **apply** pauses for approval in the `production` environment → resources provision. A plan summary lists every resource to be created.

Keyless auth

AWS via IAM role + GitHub OIDC; GCP via Workload Identity Federation; tokens are minted per-run and short-lived. No static cloud keys live in GitHub.

7 Security & compliance

Identity & access

- GitHub OIDC → AWS IAM role / GCP WIF (no static keys).
- EKS access entries map identities to cluster RBAC.
- GKE Workload Identity for pod-level GCP access.
- Production deploys require human approval (GitHub environment).

In-transit & supply chain

- Mesh-wide **mTLS** (shared CA) for service-to-service traffic.
- Reference admission model: signed images from trusted registries, PSS.
- Remote state encrypted in S3 with locking.

Known hardening items (validation environment). The bootstrap AWS key was shared during setup and must be **rotated**; the operator IAM user currently holds `AdministratorAccess` (scope down); the GKE control-plane endpoint is publicly reachable (restrict via authorized networks); and CockroachDB currently runs in insecure mode (enable TLS for production). Deploy identifiers are held as GitHub Actions secrets; for runtime app secrets a cloud-native secret manager + External Secrets is recommended.

8 Observability

Deployed

`kube-prometheus-stack` (Prometheus + Grafana + node-exporter) on EKS; GKE uses Google Managed Service for Prometheus. Istio contributes per-service golden signals (latency, errors, throughput, saturation).

Fleet view (design)

Per-cluster Prometheus federates into a single fleet-wide view; the reference control plane also exposes Prometheus metrics + an offline dashboard (\$10).

9 Data layer — CockroachDB (deployed)

Active-active *traffic* needs active-active *data*. A distributed multi-master database lets every cloud read and write and survive a region loss.

ASPECT	IMPLEMENTATION
Engine	CockroachDB <code>v24.3.5</code> as a DaemonSet (one node per K8s node) joined into one logical DB
Networking	hostNetwork — each node advertises its routable node IP (pod IPs aren't routable cross-cloud)
Connectivity	node-to-node (:26257) over the AWS→GCP HA VPN (cross_cloud module); <code>--join</code> built from live node IPs
Deployment	CI/CD via <code>database.yml</code> (discovers node IPs → applies to both clusters → <code>init</code> once)
Status	LIVE — 4 nodes across both clouds (2 EKS + 2 GKE)

Hardening. The cross-cloud VPN is up and the cluster is live in **insecure** mode for this validation; enabling TLS (`cert-manager / cockroach cert`) is the production-hardening step.

10 Reference control plane (offline model)

The repository also contains an agent-driven control-plane **reference implementation** (Python) that models the full lifecycle — reconciliation loop, provider abstraction, node agent, security gate, networking fabric, observability — and runs entirely offline with **80 passing tests**. It proves the design (including the active-active traffic model in `networking/global_lb.py` + `sim/active_active.py`) deterministically.

On the live managed clusters, its responsibilities (provision / scale / self-heal) are met by Terraform + managed node groups; the **Provider** interface is the documented seam for driving real clouds directly in a future phase. An in-cluster deployment of this control plane + dashboard is on the roadmap.

11 Operations runbook

Access

```
aws eks update-kubeconfig --name bullion-eks --region us-east-1 --alias eks
gcloud container clusters get-credentials bullion-gke --zone us-central1-a --project project-f38bfd6-70ff-4a8d-9cd
```

Deploy / change

Open a PR (auto **plan**) → review → merge → approve the gated **apply**. Platform layer: Actions → **platform** → Run workflow → approve.

Verify active-active

```
kubectl --context eks -n demo run curl --image=curlimages/curl --restart=Never --command -- sleep 3600
kubectl --context eks -n demo exec curl -c curl -- sh -c \
  "for i in 1 2 3 4 5 6 7 8; do curl -s hello.demo.svc.cluster.local; echo; done"
# expect a mix of "hello from aws" and "hello from gcp"
```

Failure handling

A cloud/endpoint failure is ejected by **outlierDetection** and traffic shifts to the survivor automatically. No manual failover is required for the traffic layer.

Teardown (stop costs)

Actions → deploy → Run workflow → action=destroy → approve

12 Testing & verification

LAYER	COVERAGE
Unit/integration (offline)	80 tests (smoke/unit/integration/chaos) — control logic, security gate, networking, active-active model
IaC	terraform fmt + validate in CI on every change
Live mesh	cross-cloud LB verified (12 AWS / 8 GCP from an EKS client)

13 Implementation journey — blockers resolved

SYMPTOM	RESOLUTION
EKS rejected Kubernetes 1.29	upgraded to 1.32
GKE Autopilot can't host self-managed Istio / DaemonSets	switched GKE to Standard
Node pool rejected GKE_METADATA	enabled Workload Identity on the cluster
istiod wouldn't schedule on medium nodes	moved to 4 vCPU/16 GB nodes
GCP free-trial IP quota (4) exceeded	private nodes + Cloud NAT
Terraform state drift from partial applies	fresh plan reconciled
EKS sidecar-injector webhook timeout	opened node SG :15017 / :15012
DestinationRule distribute+failover rejected	distribute + outlierDetection
Cross-cluster traffic stayed local	added topology.istio.io/network label

14 Current state & roadmap

CAPABILITY	STATE
EKS + GKE provisioned (IaC + CI/CD)	✓ live
Istio multi-primary mesh — clusters joined	✓ live
Active-active cross-cloud load balancing	✓ verified
North-south ingress gateways (public LB :80)	✓ live
Board game — global active-active app (GHCR → both clouds)	✓ live
Observability (Prometheus/Grafana)	✓ live
Cross-cloud VPN (AWS↔GCP)	✓ live (4 tunnels)
Central CockroachDB (multi-region, via CI/CD)	✓ live — 4 nodes
In-cluster control plane + dashboard	planned
Global DNS (single-endpoint north-south LB)	module built (optional)

15 Cost

ITEM	INDICATIVE COST
EKS control plane	~\$0.10 / hr
EKS nodes (2× m5.xlarge)	~\$0.38 / hr
GKE (Standard) + nodes	on GCP free-trial credit
LoadBalancers / NAT / (VPN when enabled)	hourly + data egress

Run the destroy pipeline when idle. Cross-cloud data egress is the main variable cost once the VPN + database are enabled.

16 Risks & recommendations

RISK	RECOMMENDATION
Broad IAM (Administrator) + shared bootstrap key	rotate the key; scope roles to least privilege
Public GKE control-plane endpoint	restrict with master authorized networks
Free-trial quotas (GCP IPs/CPU)	request uplifts before scaling / production
CockroachDB runs in insecure mode	enable TLS (cert-manager / <code>cockroach cert</code>) for production
Runtime app-secret management	adopt a cloud-native secret manager + External Secrets Operator

17 Appendices

A · Repository layout

```
iac/terraform/  EKS, GKE, global_dns, cross_cloud + S3
backend
mesh/install/  Istio multi-primary (CA, IOPs, east-west +
ingress, secrets)
mesh/apps/     global service + active-active
DestinationRule + board game
mesh/observability/ node-exporter + federated Prometheus
mesh/data/     CockroachDB multi-region (cross-cloud, via
database.yaml)
```

B · Key identifiers

AWS account	730767193290
EKS cluster	bullion-eks (us-east-1)
GCP project	project-f38bfcd6-70ff-4a8d-9cd
GKE cluster	bullion-gke (us-central1-a)
Repo	Olisaemeka111/bullion-infrastructure

```
.github/workflows/ ci · deploy · platform · database · game
control_plane/ providers/ agent/ security/
networking/ workflows/ sim/ tests/ (offline reference)
DEPLOY · CICD · INFRASTRUCTURE · ARCHITECTURE · RUNBOOK ·
TESTING
```

State bucket

fleet-mini-tfstate-59998ac4

C · Glossary

Multi-primary

An istiod control plane in each cluster.

Multi-network

Clusters on different networks joined via east-west gateways.

Active-active

All clouds serve traffic at once (vs cold standby).

OIDC

Keyless federated auth — short-lived tokens, no stored secrets.

Bullion Infrastructure — Project Documentation v1.0 · active-active multi-cloud Kubernetes · repo Olisaemeka111/bullion-infrastructure · 2026-06-23. — End of document.